



Prostate capsule segmentation from micro-ultrasound images using adaptive focal loss

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ABSTRACT

Micro-ultrasound (micro-US) is a promising imaging technique for cancer detection and computer-assisted visualization. This study investigates prostate capsule segmentation using deep learning techniques from micro-US images, addressing the challenges posed by the ambiguous boundaries of the prostate capsule. Existing methods often struggle in such cases, motivating the development of a tailored approach. This study introduces an adaptive focal loss function that dynamically emphasizes both hard and easy regions, taking into account their respective difficulty levels and annotation variability. The proposed methodology has two primary strategies: integrating a standard focal loss function as a baseline to design an adaptive focal loss function for proper prostate capsule segmentation. The focal loss baseline provides a robust foundation, incorporating class balancing and focusing on examples that are difficult to classify. The adaptive focal loss offers additional flexibility, addressing the fuzzy region of the prostate capsule and annotation variability by dilating the hard regions identified through discrepancies between expert and non-expert annotations. The proposed method dynamically adjusts the segmentation model's weights better to identify the fuzzy regions of the prostate capsule. The proposed adaptive focal loss function demonstrates superior performance, achieving a mean dice coefficient (DSC) of 0.940 and a mean Hausdorff distance (HD) of 1.949 mm in the testing dataset. These results highlight the effectiveness of integrating advanced loss functions and adaptive techniques into deep learning models. This enhances the accuracy of prostate capsule segmentation in micro-US images, offering the potential to improve clinical decision-making in prostate cancer diagnosis and treatment planning.

1. Introduction

Prostate cancer is the most commonly diagnosed cancer among men. In 2018, there were approximately 164,690 new cases and 29,430 deaths from prostate cancer in the United States (US) [1]. By 2022, the number of new cases had risen to about 268,490, making it the second leading cause of cancer-related deaths in men, with an estimated 34,500 deaths that year [2]. Some projections suggest that by 2030, there could be 1.7 million prostate cancer patients worldwide [3]. Therefore, early detection of prostate cancer is crucial for improving survival rates and quality of life for patients. Early diagnosis of clinically significant prostate cancer can increase the 5-year survival rate from 31% to nearly 100% [4]. In the US, approximately one million prostate biopsies are performed annually using transrectal ultrasound (TRUS) guidance [5]. However, because TRUS imaging has limited accuracy, many clinical guidelines recommend combining targeted biopsies with blind systematic approaches to enhance diagnostic reliability [6]. Micro-ultrasound (micro-US) is a novel 29-MHz ultrasound technology that delivers 3–4 times higher resolution

than traditional ultrasound. Several studies [7,8] suggest that it offers comparable diagnostic accuracy to MRI at a lower cost. Furthermore, its real-time visualization removes the necessity for image fusion, which is responsible for up to 50% of diagnostic errors in MRI-ultrasound fusion-guided biopsies, as observed by Williams and his colleagues [9]. Micro-US enhances the detection of clinically significant cancers and reduces the risk of disease progression at an earlier stage. Manual segmentation remains the most reliable method for accurately delineating the prostate capsule area. However, prostate segmentation on micro-US is cognitively challenging for three reasons: first, while urologists and radiologists typically use axial imaging, micro-US employs oblique planes. Second, the midline often has unclear boundaries between the prostate, bladder, and adjacent vasculature. Third, micro-US images can exhibit artifacts. Consequently, micro-US determines biopsy locations relative to the urethra (midline) rather than the capsule. Its varying shapes and sizes, and the challenges of grayscale annotation, require specialized training for surgeons. The difficulty in distinguishing the prostate capsule from surrounding structures further complicates

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segmentation, making manual processes time-consuming and impractical for large populations. While micro-US improves accuracy, many urologists still prefer MRI for treatment planning due to its ability to generate 3D cancer maps. Therefore, precise and scalable prostate capsule segmentation is crucial for accurately locating prostate cancer concerning the gland.

The proposed method in this paper introduces an adaptive weighting scheme that dynamically adjusts the weights of the proposed segmentation model based on the difficulty of regions and the model's performance during training. This innovation enhances segmentation accuracy and robustness, making the model more accurate and effective for diverse patient datasets. The proposed method optimizes training processes and ensures adaptability to various imaging conditions by improving generalizability and efficiently allocating computational resources. Additionally, the scheme mitigates the impact of annotation variability, resulting in more reliable outcomes. These advancements directly benefit clinical applications, enabling precise diagnoses, better treatment planning, and improved patient outcomes while reducing healthcare costs. The main contributions of this paper are summarized below.

1. This paper introduces an adaptive focal loss function that dynamically adjusts its weighting scheme based on the patient micro-US images' specific challenges, such as the prostate capsule fuzzy boundary and annotation variability.
2. The proposed adaptive focal loss function outperforms the existing methods, such as the standard focal loss, dice-loss function, annotation-guided binary cross entropy (AG-BCE) and standard BCE loss function by dynamically adjusting to the difficulty of regions in micro-ultrasound images. This novel approach significantly improves segmentation accuracy, achieving higher Dice coefficients (DSC) and lower Hausdorff distances (HD) compared to Jiang et al.'s method [10], especially in challenging regions with low contrast or anatomical variability.
3. The proposed method in this paper demonstrates how an adaptive loss function can improve performance in complex medical imaging tasks, highlighting a shift from static to context-aware and data-responsive approaches.

2. Related work

2.1. Deep learning models for prostate segmentation

Accurate segmentation of the prostate capsule is crucial for various clinical applications, such as diagnosing and planning treatment for prostate cancer. Technological advances have facilitated the development of reliable and precise MRI image segmentation methods. Likewise, notable advancements have been made in segmenting the prostate capsule using micro-US and TRUS imaging. Nonetheless, previous research struggled with attaining optimal segmentation accuracy and managing complex regions effectively.

Chen et al. [11] introduced a modified 3D AlexNet for automatic prostate cancer classification using magnetic resonance images. The model was simplified to reduce parameters and training time and incorporated bottleneck structures, batch normalization, and global average pooling to enhance performance. Overfitting management techniques addressed the challenge of limited dataset sizes. The model achieved 92.1% accuracy, with an AUC of 0.964, outperforming ResNet 50 and Inception-V4. However, limitations include sub-90% segmentation accuracy, algorithm optimization challenges, and better multi-modal integration to reduce mismatches and improve results.

Ghavami et al. [12] conducted a quantitative comparison of six open-source segmentation algorithms tailored for prostate magnetic resonance imaging segmentation. The evaluated models included UNet, VNet, HighRes3dNet, HolisticNet, DenseVNet, and Adapted UNet. These algorithms were chosen for their relevance and reproducibility

in clinical imaging tasks. The final segmentation was achieved by integrating the predictions from five models using a majority voting mechanism in each voxel. Performance was primarily evaluated using Dice scores, providing a reliable metric to compare segmentation accuracy across networks. The key limitations of the study included a restricted dataset size, possible overfitting due to hyperparameter optimization, and a primary focus on segmentation accuracy, which may have overlooked factors such as training and inference time. This research offers a foundational analysis of segmentation algorithms, emphasizing the importance of thorough evaluation and validation to ensure applicability in real-world clinical settings.

Guo et al. [13] proposed a modified UNet deep neural network for prostate capsule segmentation in Micro-Ultrasound images. Using a dataset of 2099 images (1296 annotated), the model achieved high Intersection over Union (IoU) scores: 95.05% (training), 93.18% (validation), and 85.14% (test). A 10-fold cross-validation yielded an average IoU of 94.25% and 9% accuracy, demonstrating robust performance. This research highlights the potential of deep learning in enhancing surgical and diagnostic precision. However, the small sample size and potential annotation biases limit generalizability. Jia et al. [14] proposed a prostate segmentation method using a coarse-to-fine strategy combining probabilistic atlas-based segmentation (via DRAMMS) and Deep Convolutional Neural Networks (DCNNs) for fine segmentation. Validated on the PROMIS12 dataset using 5-fold cross-validation on 50 samples, the method achieved high accuracy: Dice Similarity Coefficient (DSC) 0.88 ± 0.04 , Average Boundary Distance (ABD) 1.74 ± 0.42 mm, and HD (95% HD) 5.00 ± 1.25 mm, outperforming traditional methods in DSC and 95% HD. This approach highlights the effectiveness of combining atlas-based and deep-learning methods for prostate MR segmentation. Limitations include segmentation mismatches due to surrounding tissue complexities.

Karimi et al. [15] also proposed a CNN-based method for robust automatic segmentation of prostate CTV in TRUS images, overcoming challenges such as weak landmarks and artifacts. The technique employed adaptive sampling during training to focus on hard-to-segment images and leverages a CNN ensemble to estimate segmentation uncertainties. Uncertain segmentations are refined using prior shape information from a statistical shape model. This approach achieved a Dice score of $93.9 \pm 3.5\%$ and an HD of 2.7 ± 2.3 mm, outperforming competing methods and significantly reducing segmentation errors. The study demonstrates that incorporating uncertainty estimation and prior shape knowledge can enhance CNN-based medical image segmentation, particularly in challenging cases.

Peng et al. [16] introduced the H-ProSeg method, a hybrid approach for robust prostate segmentation in transrectal ultrasound (TRUS) images. It uses a small number of radiologist-defined seed points to address challenges like ambiguous boundaries and anatomical variations. The method includes three subnetworks: a principal curve-based model, a differential evolution-based artificial neural network (ANN), and an ANN subnetwork for smooth contour description. Tested on 55 brachytherapy patients, H-ProSeg achieved a DSC of 95.8%, a Jaccard coefficient of 94.3%, and an accuracy (ACC) of 95.4%, with minimal performance drop (up to 2.5%) under noise. It outperformed current techniques, offering improved prostate cancer segmentation.

In the meantime, Peng and his other colleagues [17] proposed a two-stage method named Auto-ProSeg for prostate segmentation in TRUS images. They used an attention-based U-Net for initial contours and a refined principal curve-based approach with an evolutionary neural network. This hybrid method improved performance by 6%, with more consistent results (standard deviation under 4.5%) compared to over 5% without refinement. Auto-ProSeg outperformed methods like U-Net++, which achieved a DSC of 89.6% with a standard deviation of 5.8%.

2.2. State-of-the-art methods on the public micro-ultrasound dataset

Several recent studies have advanced prostate capsule segmentation in micro-US imaging by leveraging deep learning architectures and transformer-based designs. Despite notable performance gains, these methods continue to face challenges related to boundary ambiguity, annotation variability, data efficiency, or reliance on architectural complexity, particularly in difficult anatomical regions.

Zhang et al. [18] applied a deep learning-based segmentation framework to delineate the prostate and central gland from micro-ultrasound (MUS) images. By exploiting the high spatial resolution of MUS imaging (>20 MHz), their model achieved Dice similarity coefficients (DSC) of 0.918 for the prostate and 0.833 for the central gland, with average surface-to-surface distances (SSD) of 1.176 mm and 1.795 mm, respectively. External validation on a public MUS dataset further improved prostate segmentation performance, yielding a DSC of 0.957 and a Hausdorff distance (HD) of 1.922 mm. While this method demonstrates strong region overlap, it relies primarily on architectural design and supervised learning, without explicitly addressing annotation variability or dynamically adapting to boundary uncertainty, which remain critical challenges in micro-US prostate capsule segmentation.

Jiang et al. [10] introduced MicroSegNet, a TransUNet-based model incorporating multi-scale deep supervision and an annotation-guided binary cross-entropy (AG-BCE) loss to emphasize difficult regions during training. The model achieved a DSC of 0.939 and an HD of 2.02 mm, outperforming several earlier approaches. However, the fixed weighting strategy of AG-BCE does not adapt to varying degrees of sample difficulty or prediction confidence. Additionally, the study reports limitations related to single-institution data collection, restricted representation of extracapsular extension, and persistent inaccuracies along the prostate midline—regions characterized by severe boundary ambiguity and inter-observer variability.

Building on transformer-based architectures, Al-Qurri et al. [19] proposed an enhanced segmentation framework combining a dual-encoder design (EfficientNet CNN and Swin Transformer), a Hypergraph Neural Network (HGNN) for non-pairwise feature modeling, and a Mamba-v2-based VSSD decoder for refined boundary localization. Evaluated on the public micro-US dataset, their method achieved a DSC of 0.9416 and a 95th-percentile Hausdorff Distance (HD95) of 1.93 mm, surpassing MicroSegNet and Swin-UNet. While effective, this approach introduces substantial architectural complexity and computational overhead, and it primarily improves performance through network design rather than addressing loss-level adaptation to boundary uncertainty or annotation inconsistency.

Radcliffe et al. [20] addressed data efficiency and robustness by adapting the promptable foundation model SAM1 for prostate segmentation in rotational micro-ultrasound scans. Their proposed SliceTrack-SAM model employs novel prompting strategies to enhance 3D contextual awareness, achieving a Dice coefficient of 94.0% and a Hausdorff distance of 1.76 mm on the MicroSegNet dataset. Despite strong performance, the method depends on prompt engineering and model adaptation strategies that may limit generalizability and increase deployment complexity, particularly in fully automated clinical workflows.

Huang et al. [21] proposed HEFFLPNet, a Transformer-based UNet built upon MicroSegNet, incorporating hierarchical feature fusion modules (TriCAFE, MSPMA, and UpFA) and deep supervision using AG-BCE loss. The method achieved a DSC of 0.938 and an HD95 of 2.12 mm on the micro-US dataset, as well as strong generalization to the CCH-TRUSPS dataset. However, similar to other AG-BCE-based approaches, the loss formulation remains static and does not dynamically adjust to sample difficulty or annotation variability, potentially limiting performance in regions with severe boundary ambiguity.

Finally, Hung et al. [22] introduced a minimally user-guided 3D micro-ultrasound segmentation framework based on ScribblePrompt U-Net with bidirectional mask propagation. Requiring user clicks only on boundary slices, the method achieves a DSC of 94.0% and an HD95 of

2.86 mm on a public micro-US dataset. Although effective in reducing annotation effort, the reliance on user interaction and 3D propagation introduces additional complexity and may restrict scalability in time-sensitive or fully automated clinical settings.

2.3. Adaptive loss functions in image segmentation

Several boundary-aware segmentation losses based on the distance transform map have been proposed. Zhu et al. [23] introduced a boundary-weighted segmentation loss (BWSL) function to make the segmentation network more sensitive to boundaries during segmentation. Kervadec et al. [24] proposed a boundary loss formulated as a distance metric focusing on contours rather than regions. This approach addresses unbalanced segmentation challenges by emphasizing integrals over boundaries instead of regions. Karimi & Salcudean [25] proposed three loss functions based on the distance transform of the segmentation boundary, aimed at minimizing the HD in medical image segmentation. Moreover, several researchers explored the adaptive loss function. For instance, Lin et al. [26] developed the focal loss to address the class imbalance in object detection tasks, which has been widely adopted in medical image segmentation for its effectiveness in challenging regions. However, its application to micro-ultrasound images may require further customization to handle the high resolution and calcification artifacts characteristic of these images, which can obscure boundary details and complicate segmentation. Kamnitsas et al. [27] proposed an adaptive deep learning model for brain tumor segmentation, achieving significant improvements in accuracy and robustness by using an adaptive loss function that adjusts during training based on performance. While this approach demonstrated strong potential, its direct application to prostate segmentation in micro-US images may require adaptations to address the unique imaging characteristics and clinical requirements, such as handling high-frequency noise and artifacts specific to micro-US technology.

The primary objective of this work is to enhance the accuracy and robustness of prostate capsule segmentation in micro-US images by directly addressing the fundamental limitations of existing approaches. We propose a framework that integrates multi-scale deep supervision with an adaptive focal loss, enabling effective learning of fine-grained features across scales while maintaining robustness to image variability and annotation uncertainty. Unlike prior methods that increase architectural complexity or require user interaction, the proposed approach targets the core challenges of micro-US segmentation at the loss-function level.

The adaptive focal loss dynamically modulates learning according to sample difficulty and annotation variability, emphasizing hard-to-segment regions while preserving overall segmentation stability. Although the proposed model has marginally higher representational capacity, the observed performance improvements are primarily driven by this adaptive loss formulation rather than architectural modifications. This design yields improved regional overlap and boundary precision without increasing inference complexity or computational overhead. The key innovation lies in the loss function's adaptability, which allows the model to effectively manage the noise and heterogeneity inherent in micro-US imaging. Consequently, the framework delivers consistent boundary delineation in challenging regions while remaining suitable for real-time clinical deployment. Furthermore, the approach is flexible and readily extensible to other medical imaging tasks facing similar segmentation challenges, reinforcing its practical and translational value.

3. Methods

The proposed method in this section outlines a comprehensive approach for prostate capsule segmentation from micro-US images, leveraging an adaptive focal loss function to address class imbalance and improve segmentation accuracy. This section offers a comprehensive summary of the details of the proposed method.

3.1. Algorithm: Adaptive focal loss for prostate capsule segmentation from micro-ultrasound images

Algorithm 1 Adaptive Focal Loss for Prostate Capsule Segmentation from Micro-Ultrasound Images

Require: Micro-ultrasound image I

Ensure: Prostate capsule segmentation mask S

- 1: **Preprocessing:**
 - a. Load input image I .
 - b. Normalize pixel intensity values to $[0, 1]$.
 - c. Apply augmentation (rotation, scaling, flipping).
- 2: **Model Initialization:**
 - a. Initialize encoder–decoder segmentation model.
 - b. Integrate adaptive focal loss for class imbalance.
 - c. Load pre-trained weights or initialize parameters.
- 3: **Feature Extraction:**
 - a. Apply dilated convolutions to expand the receptive field while preserving I 's spatial resolution.
 - b. Encode I with a transformer to obtain multiscale features F .
- 4: **Adaptive Attention Mechanism:**
 - a. Apply channel and spatial attention to refine features.
 - b. Generate refined feature maps F_{refined} from the upsampled features.
- 5: **Decoder and Segmentation:**
 - a. Decode F_{refined} using skip connections.
 - b. Upsample to match the resolution of I .
 - c. Generate raw mask S_{raw} .
- 6: **Loss Function and Optimization:**
 - a. Define adaptive focal loss:
$$L_{\text{adaptive}} = -\alpha_t(1 - p_t)^\gamma \log(p_t),$$
 - where α_t balances class weights, γ emphasizes hard pixels.
 - b. Combine adaptive focal loss with Dice loss.
 - c. Optimize using Adam with a learning rate scheduler.
- 7: **Training and Segmentation:**
 - a. Decode F_{refined} (as described in Step 5) to perform segmentation.
 - b. Train the model on a labeled dataset using the combined loss (Step 6).
 - c. Monitor DSC and HD on the validation set.
- 8: **Inference:**
 - a. Predict segmentation mask S for I .
 - b. Apply postprocessing (morphological operations).
 - c. Evaluate S using DSC and HD.

3.2. Proposed micro-ultrasound segmentation model

Fig. 1 illustrates the architecture of the proposed segmentation model, which extends the TransUNet framework, a transformer-based approach specifically designed for medical image segmentation. Two critical modifications were implemented to improve the performance of the TransUNet model. First, a multi-scale deep supervision module was integrated to capture global contextual dependencies while preserving local information. Second, adaptive focal loss was incorporated into the training process to mitigate class imbalance by dynamically modulating the loss contribution of individual pixels. Collectively, these enhancements significantly improved the model's segmentation performance.

3.2.1. TransUNet model architecture

The Vision Transformer (ViT) [28] is a transformer-based architecture originally designed for image classification. ViT processes images as tokenized patches and uses self-attention mechanisms to capture

global contextual information effectively [29]. TransUNet is the first transformer-based model for medical image segmentation, integrating the strengths of ViT and UNet architectures. It combines global context modeling with high-resolution spatial details through a hybrid CNN-transformer design, enabling precise localization by merging self-attentive features with high-resolution CNN features. This combination achieves superior performance compared to traditional CNN-based segmentation models. The main components of TransUNet are described below.

Convolutional stem. A convolutional stem layer extracts low-level features from the input image, utilizing dilated convolutions in the encoder block to preserve spatial resolution and expand receptive fields. This approach is crucial for accurate boundary delineation in prostate capsule segmentation.

Image sequentialization. Input features x from the convolutional stem are tokenized into a sequential representation of flattened 2D patches, denoted $x_i^p \in \mathbb{R}^{P^2 \cdot C}$ for $i = 1, \dots, N$. Each patch has dimensions $P \times P$, and the total number of patches is $N = HW/P^2$, where H and W are the height and width of the input image. This process converts the input into a fixed-length sequence.

Patch embedding. The patch embedding layer divides the feature map into fixed-size patches x_p and projects them into a D -dimensional embedding space using a linear transformation. Positional embeddings are added to the patch embeddings to encode spatial information (Eq. (1)):

$$z^0 = [x_1^p E; x_2^p E; \dots; x_N^p E] + E_{\text{pos}}, \quad (1)$$

where $E \in \mathbb{R}^{(P^2 \cdot C) \times D}$ is the patch embedding matrix, and $E_{\text{pos}} \in \mathbb{R}^{N \times D}$ represents the positional embeddings.

Transformer encoder. The transformer encoder comprises L layers, each featuring multi-head self-attention (MHSA) and feed-forward networks (FFN), with residual connections and layer normalization. Patch embeddings are iteratively refined by attending to relevant patches. For the ℓ th layer, the output is (Eqs. (2) & (3)):

$$z'_\ell = \text{MHSA}(\text{LN}(z_{\ell-1})) + z_{\ell-1}, \quad (2)$$

$$z_\ell = \text{FFN}(\text{LN}(z'_\ell)) + z'_\ell. \quad (3)$$

where $\text{LN}(\cdot)$ is the layer normalization operator, and z_L is the final encoded representation of the image.

UNet decoder. The UNet decoder consists of up-sampling blocks with convolutional layers, up-sampling layers, and skip connections to the convolutional stem layers. Patch embeddings are up-scaled to the original image resolution, and skip connections integrate high-resolution features from the stem with high-level features from the transformer encoder. The final output is a segmentation map, labeling each pixel in the input image.

3.2.2. Multi-scale deep supervision

The TransUNet model is a baseline for reducing discrepancies between predicted segmentation maps and ground truth. However, it may face challenges related to input image shape, size, and appearance variations. Multi-scale deep supervision is implemented to address these issues, integrating information from multiple scales during the segmentation process [30]. Each network pathway emphasizes features at a specific scale, enabling the model to combine local and global information.

To enhance performance, 1×1 convolutional layers with Sigmoid activation are applied to intermediate features at three scales ($1/2$, $1/4$, and $1/8$). This generates prostate segmentation maps at varying resolutions, where lower resolutions capture global context and higher resolutions focus on local details. For accurate predictions at all scales, ground truth segmentation maps are downsampled, and the differences between each prediction and its corresponding downsampled ground truth are incorporated into the loss function. By supervising predictions across multiple scales, the model effectively integrates features at different levels, capturing both high-level and detailed contextual information.

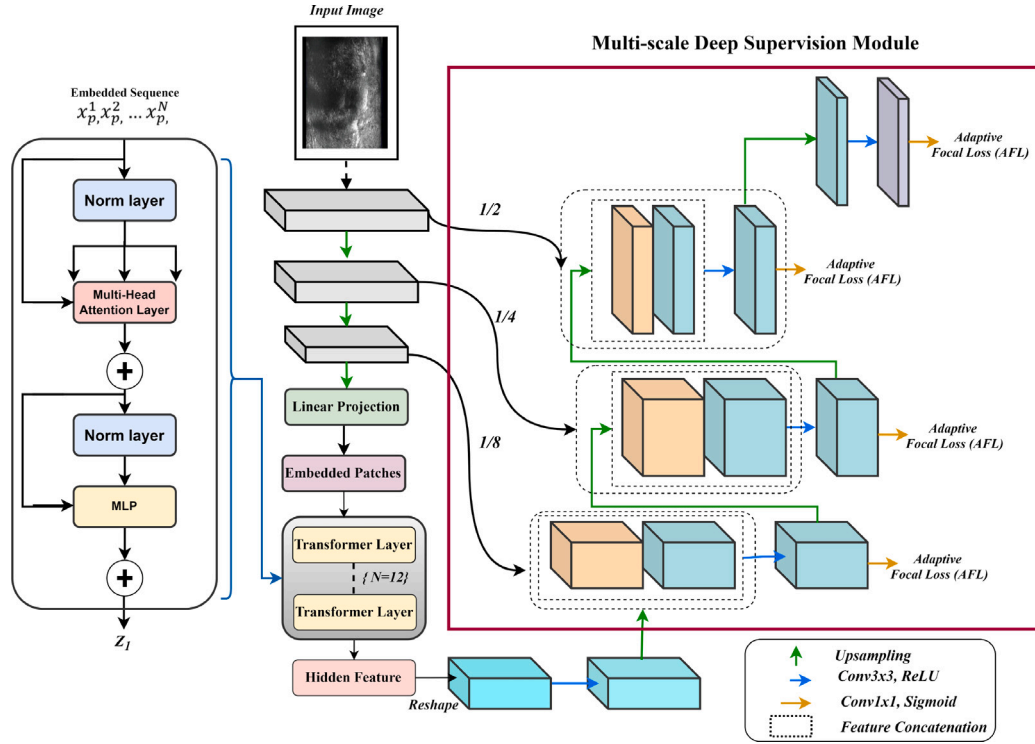


Fig. 1. Overview of the proposed segmentation model. The model leverages the TransUNet architecture, with multi-scale deep supervision in intermediate layers for enhanced global context and the proposed adaptive focal loss in the output layer to improve accuracy, particularly in challenging regions.

3.3. Loss function design

3.3.1. Proposed adaptive focal loss

In this subsection, we first analyze the limitations of existing loss functions for medical image segmentation, with particular emphasis on class imbalance, sample difficulty, and annotation uncertainty, and then introduce the proposed Adaptive Focal Loss. The standard binary cross-entropy (BCE) loss is widely used in segmentation tasks; however, it exhibits notable limitations when applied to highly imbalanced medical imaging problems. BCE optimizes pixel-wise classification accuracy while treating all pixels independently, which often leads to poor alignment with overlap-based evaluation metrics such as the Dice coefficient [31]. Consequently, BCE frequently underperforms when segmenting small structures, motivating the adoption of more region-aware and adaptive loss functions.

To address some of these shortcomings, the AG-BCE loss function [10] assigns different weights to “hard” and “easy” regions based on annotations. While effective to some extent, AG-BCE applies uniform weighting within these regions, disregarding inherent variations in sample difficulty and annotation uncertainty. This fixed weighting strategy can limit the model’s learning capacity, particularly in complex datasets characterized by high variability or ambiguous regions. In contrast, adaptive weighting mechanisms can enhance learning by allowing the model to dynamically focus on more challenging areas. Similarly, despite its widespread use, Dice loss also presents limitations in highly imbalanced segmentation settings. It is sensitive to class volume and can become unstable when dealing with small target structures, leading to suboptimal learning dynamics [32]. These observations motivate the development of generalized or adaptive formulations that explicitly address class imbalance.

Incorporating sample difficulty into the loss function is therefore crucial, as different samples present varying levels of challenge. Samples located near decision boundaries or exhibiting high prediction uncertainty are inherently more difficult to classify. Accounting for this difficulty within the loss function ensures increased attention to

such cases, ultimately improving overall performance. This principle aligns with the work of Lin et al. [26], who introduced Focal Loss to emphasize hard-to-classify examples by scaling loss contributions based on model confidence.

Another important challenge arises from annotation variability, which often reflects disagreement or uncertainty among annotators. Regions with high annotation variability require different learning strategies compared to consistently labeled areas. Explicitly modeling this variability allows the network to focus more effectively on uncertain regions, consistent with the findings of Kendall et al. [33], who explored the role of uncertainty in loss functions for tasks such as semantic segmentation and object detection.

Motivated by the above considerations, we now detail the formulation of the proposed Adaptive Focal Loss. To achieve adaptivity, both sample difficulty and annotation variability are incorporated into a dynamic gamma factor that regulates the influence of hard and easy regions. This dynamic formulation enables the loss to adjust its behavior on a per-sample basis, leading to more robust and effective learning in highly imbalanced settings. Related approaches, such as Dynamic Loss Weighting (DWL) [34], similarly emphasize adapting loss functions according to task or sample importance.

By synthesizing these concepts, the proposed Adaptive Focal Loss addresses inherent data challenges and enhances performance in complex and uncertain environments. The loss dynamically adjusts the gamma factor based on real-time estimates of sample difficulty and annotation variability, enabling it to outperform traditional approaches such as AG-BCE, particularly in datasets with high complexity and uncertainty. The pseudocode for the proposed adaptive focal loss is provided below.

4. Experimental setup

4.1. Dataset acquisition and preprocessing

This study utilizes the Micro-ultrasound Prostate Capsule Segmentation dataset [10], which consists of micro-ultrasound images from

Algorithm 2 Pseudocode for the Adaptive Focal Loss Function

Require: Expert ground-truth mask y_{true} , predicted probability map $y_{\text{pred}} \in [0, 1]$ (after sigmoid), non-expert annotation mask y_{std} , dilation kernel size ks , and small constant ϵ

Ensure: Final loss $LOSS$

- 1: **Step 1: Calculate the number of pixels**
 $N_{\text{pixels}} = \text{height} \times \text{width} \times \text{channels}$ $\triangleright$ Entire image (foreground + background).
- 2: **Step 2: Identify hard regions**
 $y_{\text{true_np}} \leftarrow \text{to_numpy}(y_{\text{true}})$ $\triangleright$ Expert ground-truth mask in NumPy.
 $y_{\text{std_np}} \leftarrow \text{to_numpy}(y_{\text{std}})$ $\triangleright$ Non-expert annotation mask / variability map in NumPy.
 $\text{hard_np} = y_{\text{true_np}} \oplus y_{\text{std_np}}$ $\triangleright$ Bitwise XOR to identify disagreement regions.
- 3: **Step 3: Perform dilation**
 $\text{hard_dilated} = \text{dilate}(\text{hard_np}, \text{kernel}(ks \times ks))$ $\triangleright$ In experiments we set $ks = 3$ (i.e., a 3×3 kernel).
- 4: **Step 4: Define easy regions**
 $\text{easy} = 1 - \text{hard_dilated}$
- 5: **Step 5: Clamp predicted probabilities for numerical stability**
 $y_{\text{pred}} \leftarrow \text{clip}(y_{\text{pred}}, \epsilon, 1 - \epsilon)$
- 6: **Step 6: Set focal loss parameter**
 $\beta = 0.5$ $\triangleright$ Class-balancing factor.
- 7: **Step 7: Calculate sample difficulty and annotation variability**
 $\text{sample_difficulty} = 1.0 - \text{mean}(y_{\text{pred}})$
 $\text{annotation_variability} = \text{mean}(y_{\text{std}})$
- 8: **Step 8: Compute gamma factor**
 $\gamma = \text{sample_difficulty} + \text{annotation_variability}$
- 9: **Step 9: Compute focal loss**
 $p_t = y_{\text{pred}} \cdot y_{\text{true}} + (1 - y_{\text{pred}}) \cdot (1 - y_{\text{true}})$
 $\text{focal_loss} = -\beta \cdot (1 - p_t)^\gamma \cdot \log(p_t + \epsilon)$
- 10: **Step 10: Compute hard and easy losses**
 $\text{hard_loss} = \sum (\text{focal_loss} \times \text{hard_dilated})$
 $\text{easy_loss} = \sum (\text{focal_loss} \times \text{easy})$
- 11: **Step 11: Adjust hard and easy losses**
 $\text{weighted_hard_loss} = \gamma \cdot \text{hard_loss}$
 $\text{weighted_easy_loss} = \frac{1}{\gamma} \cdot \text{easy_loss}$
- 12: **Step 12: Compute final loss**
 $LOSS = \frac{\text{weighted_easy_loss} + \text{weighted_hard_loss}}{N_{\text{pixels}}}$ $\triangleright$ Final loss averaged over all pixels.
- 13: **Step 13: Return final loss**
return $LOSS$

75 patients. The training set comprises 2750 images from 55 patients, while the test set consists of 600 images from 20 patients. Each patient's dataset includes approximately 200–300 micro-ultrasound images acquired in a pseudo-sagittal plane. These test cases represent independent evaluations for each test patient, with no overlap between the training and testing datasets. This ensures that the model's evaluation on the test set is independent of the training data, providing a generalizable measure of the model's performance. To maintain consistency in scale and size, the images are normalized to an intensity range of $[0, 1]$ and resized to 224×224 pixels for neural network training. In addition, various data augmentation techniques, such as random rotations, flips, and intensity variations, are applied to the training images to improve generalization. All images are annotated by both experts and non-experts, with the annotations reviewed and refined to ensure high-quality ground truth. Discrepancies in annotations are used to identify challenging regions, which are incorporated into the AG-BCE loss function during training. Fig. 2 illustrates the sample images of the dataset.

4.2. Segmenting challenging regions in prostate capsule imaging

Segmentation of the prostate capsule in micro-US images presents unique challenges due to several factors. For instance, one primary issue is the ambiguity of boundaries at the prostate-bladder interface, which often leads to variability among expert annotations. Additionally, bright calcification spots within the prostate can obscure the true outline of the capsule, further complicating the segmentation process. Anatomical variations at the prostate-vasculature junctions also contribute to inconsistencies between patients, making it harder to achieve precise segmentation. These issues lead to variability in segmentation performance, which can be categorized into hard and easy regions [10].

1. **Boundary Ambiguity:** The prostate-midline region, where the prostate and bladder merge, is difficult to segment due to indistinct boundaries. This causes significant annotation variability, with experts differing in their delineation of the prostate capsule. To tackle this, the adaptive focal loss function focuses on these challenging regions, emphasizing errors in uncertain boundary areas during training.
2. **Artifacts:** Artifacts such as calcifications and speckle noise complicate segmentation by distorting the tissue boundaries. These artifacts, particularly calcifications that appear as bright spots, confuse the model, making it harder to accurately delineate the prostate capsule. The model addresses this challenge by assigning higher penalties to errors in artifact-prone areas using the adaptive focal loss.
3. **Anatomical Complexity:** Complex anatomical structures, especially at the junction of the prostate and surrounding vasculature, vary across patients, leading to inconsistent segmentation. These regions are trained using a multi-scale deep supervision approach to ensure both global context and local details are considered, improving segmentation accuracy.

Moreover, easy regions have clearly defined boundaries, such as regions far from the bladder or those without artifacts. These regions are easier to segment and are given lower weight in the adaptive focal loss function, allowing the model to focus more on challenging areas. To improve segmentation accuracy in these challenging regions, this work introduces an adaptive focal loss function that adjusts the focus during training based on the difficulty of the regions and variability in annotations. This ensures that the model prioritizes learning in areas with higher segmentation uncertainty, ultimately improving performance, especially in complex regions.

4.3. Loss function assumptions and design considerations

The focal loss function addresses imbalanced datasets and challenging regions by dynamically adjusting the contribution of each example to the overall loss. This approach prioritizes harder examples while reducing the impact of easier ones, enhancing the model's generalization in complex scenarios. The computations for both the standard focal loss (`pyT_focal_loss`) and the proposed adaptive focal loss are based on the focal loss formulation (Eq. (4)):

$$\text{focal_loss}(y_{\text{true}}, y_{\text{pred}}) = -\beta \cdot (1 - p_t)^\gamma \cdot \log(p_t + \epsilon), \quad (4)$$

where y_{true} is the expert ground-truth segmentation mask and $y_{\text{pred}} \in [0, 1]$ is the predicted probability map (after sigmoid activation). The term p_t is the predicted probability assigned to the true class, computed as (Eq. (5)):

$$p_t = y_{\text{pred}} \cdot y_{\text{true}} + (1 - y_{\text{pred}}) \cdot (1 - y_{\text{true}}). \quad (5)$$

In practice, we use ϵ to prevent numerical issues (e.g., $\log(0)$) by clamping y_{pred} to $[\epsilon, 1 - \epsilon]$ before computing p_t . We set the class-balancing factor $\beta = 0.5$.

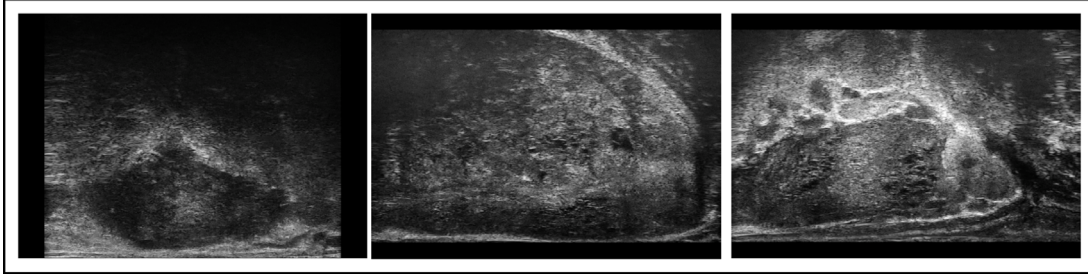


Fig. 2. Sample images of the dataset.

For the proposed Adaptive Focal Loss, the focusing factor is computed dynamically per sample as (Eq. (6)):

$$\gamma = (1 - \text{mean}(y_{\text{pred}})) + \text{mean}(y_{\text{std}}), \quad (6)$$

where y_{std} denotes the non-expert annotation mask (or annotation-variability map). Thus, $\text{mean}(y_{\text{std}})$ serves as a proxy for annotation variability by measuring the proportion of pixels marked as uncertain/discrepant across the image. The parameter γ controls the focus on hard versus easy regions, while ϵ avoids errors due to logarithms of zero. This loss function helps the model emphasize challenging regions, which is especially beneficial for imbalanced datasets or tasks with complex boundaries. The baseline focal loss implementation `pyT_focal_loss` is computed from logits and applies sigmoid internally, whereas our Adaptive Focal Loss uses sigmoid probabilities $y_{\text{pred}} \in [0, 1]$ as input.

4.3.1. Bias considerations

The focal loss functions aim to handle regions of varying difficulty with minimal bias. For harder regions, where predictions are uncertain or near decision boundaries, the $(1 - p_t)^\gamma$ term amplifies the loss, drawing the model's attention. In contrast, the loss for easier regions is naturally smaller, enabling the model to concentrate less on well-classified examples. Adaptive focal loss additionally adjusts for annotation variability, dynamically modifying the γ factor to reflect the unique challenges of each sample.

4.3.2. Dataset characteristics

The performance of these loss methods depends on dataset characteristics, such as class imbalance or inconsistent labels. The standard focal loss addresses class imbalance by emphasizing hard examples, while adaptive focal loss incorporates annotation variability to handle inconsistently labeled regions more effectively.

4.3.3. Empirical support for loss design assumptions

The design assumptions underlying the proposed Adaptive Focal Loss are empirically supported through systematic comparisons with four commonly used loss functions: PyTorch Focal Loss, annotation-guided binary cross-entropy (AG-BCE), Dice Loss, and standard BCE. These comparisons evaluate the effectiveness of adaptive weighting in handling class imbalance, boundary ambiguity, and annotation variability in prostate capsule segmentation from micro-ultrasound images. The results demonstrate that incorporating dynamic, confidence-aware modulation leads to improved learning behavior and segmentation performance for the dataset considered in this study. Detailed quantitative and qualitative comparisons are presented in the subsequent Results and Discussion section.

5. Results and discussion

5.1. Evaluation metrics

The DSC was employed to evaluate the overlap between the predicted prostate segmentation (p) and the ground truth segmentation (g)

Table 1

Comparison of loss values across epochs for different loss functions during training.

Epoch	Adaptive Focal	PyTorch Focal	AG BCE	Dice Loss	BCE Loss
1	0.212186	0.152941	0.106898	0.185000	0.198000
2	0.101646	0.134336	0.059834	0.142000	0.158000
3	0.080325	0.131814	0.050836	0.120000	0.138000
4	0.070172	0.130542	0.044332	0.108000	0.125000
5	0.061344	0.129793	0.040449	0.098000	0.115000
6	0.056536	0.129307	0.037647	0.090000	0.108000
7	0.050847	0.128894	0.035039	0.083000	0.102000
8	0.046570	0.128493	0.032370	0.077000	0.097000
9	0.043747	0.128443	0.030755	0.073000	0.093000
10	0.040419	0.128173	0.029263	0.069000	0.089000

under a setup similar to the original experiment. It is defined as (Eq. (7)):

$$\text{DSC}(g, p) = \frac{2 \times |g \cap p|}{|g| + |p|} \quad (7)$$

Here, $|g|$ and $|p|$ denote the number of positive pixels in the ground truth and predicted segmentation, respectively, while $g \cap p$ represents their intersection. The DSC ranges from 0 to 1, where higher values indicate greater overlap between the segmentations.

Additionally, the Hausdorff Distance (HD) was used to quantify the maximum distance between the boundaries of g and p . The HD is given by (Eq. (8)):

$$\text{HD} = \max \left\{ \sup_{G \in \mathcal{G}} \inf_{P \in \mathcal{P}} d(G, P), \sup_{P \in \mathcal{P}} \inf_{G \in \mathcal{G}} d(G, P) \right\} \quad (8)$$

To minimize the influence of small outliers, the 95th percentile of these distances (HD95) was adopted instead of the maximum distance.

5.2. Loss curve during training

Table 1 summarizes the loss values obtained over ten training epochs using five different loss functions: standard focal loss (i.e., `pyT_focal_loss`), `adaptive_focal_loss`, `AG_BCE_loss`, Dice loss, and BCE loss. A clear and consistent reduction in loss is observed for all methods as training progresses. The `adaptive_focal_loss` shows a pronounced decrease in loss, dropping from 0.212186 at the first epoch to 0.040419 at the final epoch. This substantial reduction reflects improved convergence behavior compared to the standard focal loss and indicates the benefit of dynamically adjusting the focusing mechanism to better emphasize difficult samples during training. In comparison, the `AG_BCE_loss` consistently produces the lowest loss values across most epochs, decreasing from 0.106898 to 0.029263, which suggests strong optimization capability in terms of absolute loss minimization. The standard focal loss exhibits a slower and more limited reduction, from 0.152941 to 0.128173, likely due to its fixed focusing parameter.

Fig. 3 illustrates the loss trends observed during training. The `adaptive_focal_loss` converges more rapidly and steadily than

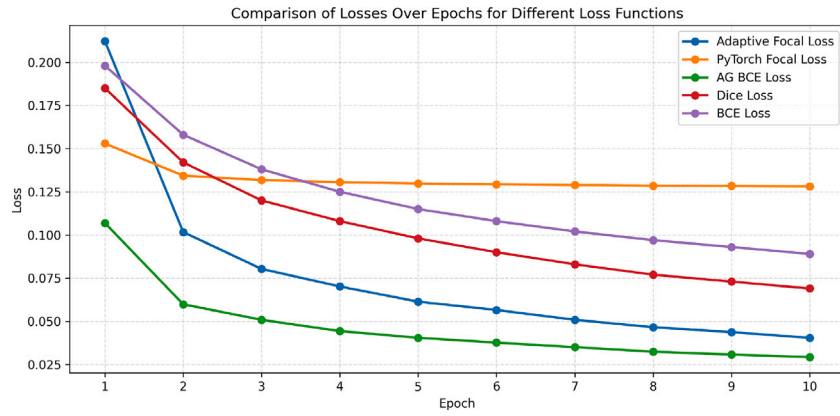


Fig. 3. Loss curve during training. The figure shows the evolution of loss values for all five loss functions across epochs, allowing comparison of their convergence behavior and training stability.

Table 2

Comparison of performance metrics across 20 test cases for different loss functions. The table reports mean Dice and mean HD95 values for each test case and the overall average for five loss functions: Adaptive Focal Loss, pyTorch Focal Loss, AG BCE Loss, Dice Loss Function, and BCE Loss.

Idx	Adaptive Focal		AG BCE		pyTorch Focal		Dice Loss		BCE	
	Dice	HD95	Dice	HD95	Dice	HD95	Dice	HD95	Dice	HD95
1	0.9569	1.9925	0.9525	2.6741	0.9317	3.0807	0.9078	3.2696	0.9210	2.1200
2	0.9254	2.3482	0.9118	3.1074	0.9396	2.1233	0.8450	4.1396	0.8720	3.0500
3	0.9576	1.1166	0.9499	1.1531	0.9160	1.7192	0.9218	1.4066	0.9030	2.3400
4	0.9421	1.8843	0.9417	1.9182	0.9046	2.8805	0.8983	2.8024	0.8890	2.7100
5	0.9381	1.5865	0.9336	1.7901	0.9133	2.2564	0.9012	2.0801	0.9150	2.1800
6	0.9344	2.6164	0.9395	2.4522	0.9004	3.1953	0.8945	3.0100	0.8610	3.2200
7	0.9554	1.4290	0.9606	1.2860	0.9408	1.7067	0.9194	1.8335	0.8970	2.5600
8	0.9577	1.2766	0.9613	1.1483	0.9284	1.9655	0.9131	1.8774	0.8830	2.8400
9	0.8939	2.6147	0.8525	3.3892	0.8445	3.1875	0.8353	3.8932	0.8690	3.1100
10	0.9550	0.9854	0.9479	1.1115	0.9166	1.6644	0.8940	1.6139	0.8920	2.6300
11	0.9426	2.1823	0.9436	2.3633	0.9194	2.9209	0.9157	2.4257	0.9050	2.2900
12	0.9150	2.4519	0.9244	2.0885	0.8860	2.9951	0.8409	3.8509	0.8740	2.9800
13	0.9244	2.0687	0.9374	1.7690	0.8987	2.4553	0.8908	2.5566	0.8880	2.7400
14	0.9403	2.1641	0.9458	2.2211	0.8905	3.0592	0.8925	2.9285	0.9010	2.4100
15	0.9539	1.5215	0.9569	1.4332	0.9435	1.7696	0.9187	1.7820	0.8790	2.8700
16	0.9341	2.1774	0.9457	1.9767	0.9050	2.5957	0.8900	3.1779	0.8860	2.7900
17	0.9467	2.4147	0.9488	2.2691	0.9168	3.9691	0.9131	2.3252	0.8930	2.6000
18	0.9547	1.6909	0.9541	1.6121	0.9299	2.2262	0.9035	2.5497	0.8670	3.1800
19	0.9542	1.4059	0.9510	1.3725	0.9134	1.8777	0.9155	2.0061	0.8980	2.5200
20	0.9178	3.0461	0.9364	2.8758	0.9403	2.2969	0.8927	4.0054	0.8810	2.9100
Avg	0.9400	1.9487	0.9398	2.0006	0.9140	2.4973	0.8952	2.6767	0.8887	2.7025

the standard focal loss, confirming the advantage of its adaptive weighting strategy in focusing on harder samples as training progresses. At the same time, the AG_BCE_loss maintains consistently lower loss values throughout training, which explains its superior numerical performance in terms of loss minimization. Dice loss and BCE loss follow similar decreasing patterns but remain higher than both adaptive focal and AG-BCE losses across all epochs. Overall, these results indicate that while AG-BCE achieves the lowest absolute loss values, the adaptive focal loss offers a meaningful improvement over traditional focal loss by enhancing convergence behavior, stability, and robustness during training.

5.3. Quantitative comparison of loss functions

Table 2 reports the segmentation performance obtained using five different loss functions across 20 independent test cases, evaluated using the Dice Similarity Coefficient (DSC) and the 95th percentile Hausdorff Distance (HD95). For each test case, DSC and HD95 are computed independently, and the final results represent the mean values averaged over all 20 cases, providing a robust overall comparison of model performance.

Among all evaluated loss functions, the proposed **Adaptive Focal Loss** consistently achieves the best overall performance. It yields the highest average DSC of **0.9400** while simultaneously producing the lowest average HD95 of **1.9487 mm**, indicating superior overlap accuracy and more precise boundary localization. Compared to the standard PyTorch Focal Loss, the Adaptive Focal Loss improves DSC from 0.9140 to 0.9400 and reduces HD95 from 2.4973 mm to 1.9487 mm. These improvements, while numerically moderate, are practically meaningful in medical image segmentation, where small boundary errors can lead to clinically significant differences.

The performance gains of the Adaptive Focal Loss stem from its ability to dynamically adapt the loss contribution based on prediction confidence, sample difficulty, and annotation variability. By emphasizing hard-to-segment regions and down-weighting well-classified pixels, the loss function encourages more accurate learning in complex anatomical structures and ambiguous boundary regions. This adaptive behavior leads to improved robustness against outliers and enhances boundary consistency across test cases.

The AG-BCE Loss also demonstrates strong performance, achieving an average DSC of 0.9398 and an HD95 of 2.0006 mm, outperforming the PyTorch Focal Loss. However, it lacks an explicit confidence-aware modulation mechanism, which limits its ability to fully adapt to varying

prediction uncertainty. As a result, its overall performance remains slightly inferior to that of the Adaptive Focal Loss. In contrast, the Dice Loss and standard BCE Loss show noticeably lower DSC values and higher HD95 distances, reflecting reduced boundary accuracy and segmentation consistency. Overall, the results in Table 2 highlight the effectiveness of incorporating adaptive, confidence-aware mechanisms into the loss formulation, with the Adaptive Focal Loss delivering the most accurate and reliable segmentation performance across all evaluated metrics.

5.4. Training setup and performance evaluation

This section evaluates the proposed framework in terms of training setup, computational efficiency, and segmentation performance. We describe the experimental configuration for reproducibility, analyze the computational impact of different loss functions on training and inference, and present qualitative visual comparisons to assess segmentation accuracy in challenging prostate boundary regions in micro-ultrasound images.

5.4.1. Computational cost and training configuration

All experiments were conducted on a high-performance server equipped with NVIDIA A100 GPUs (80 GB memory). Python 3.10.6 and PyTorch 1.12.1 were used for all experiments. Input images were resized to 224×224 and normalized to the range $[0, 1]$. During training, we employed a patch size of 16, a batch size of 8, a learning rate of 0.01 with momentum 0.9, and a weight decay of 1×10^{-4} . These hyperparameters were selected through empirical testing and applied uniformly across all models. All models were trained for 10 epochs to mitigate overfitting, and the training procedure was repeated eight times; the final Dice coefficient and Hausdorff distance were reported as the average across the eight runs.

To quantify the computational overhead introduced by different loss functions, we recorded the number of trainable parameters, FLOPs, training time, and inference latency, as summarized in Table 3. Since the loss function affects only the optimization objective and does not alter the network architecture, the number of parameters (105.28 M) and FLOPs (49.37 G) remain identical across Adaptive Focal Loss, AG-BCE Loss, PyTorch Focal Loss, Dice Loss, and BCE Loss. Similarly, inference time remains unchanged across all loss functions, with an average latency of approximately 23.0 ± 0.3 ms, indicating that the proposed losses do not affect deployment efficiency.

Differences in computational cost are observed primarily during training. Based on the epoch timing extracted from the `tqdm` training logs, the Adaptive Focal Loss required an average of 123 s per epoch (approximately 2.05 min), corresponding to a total training time of approximately 20.5 min for 10 epochs. In comparison, AG-BCE Loss, PyTorch Focal Loss, Dice Loss, and BCE Loss required 118 s, 116 s, 120 s, and 117 s per epoch, respectively. These minor timing variations are attributed to the computational complexity of each loss formulation, where the Adaptive Focal and Dice Losses incur slightly higher backpropagation costs due to additional gradient weighting and normalization operations.

5.4.2. Model performance

Based on the quantitative evaluation presented in the previous section Table 2, the standard BCE and Dice loss functions exhibited consistently inferior performance in terms of both region overlap and boundary accuracy for prostate capsule segmentation from micro-ultrasound images. In particular, these losses struggled to handle severe class imbalance, boundary ambiguity, and annotation variability, resulting in lower Dice Similarity Coefficients and higher Hausdorff distances compared to focal-based formulations. Consequently, for qualitative visualization and detailed boundary analysis, we focus on three representative and competitive loss functions: the proposed Adaptive Focal Loss, AG-BCE, and the standard PyTorch Focal Loss. Fig. 4 presents

Table 3

Computational cost of the TransUNet model under different loss functions. The loss function does not alter the network architecture; therefore, the number of parameters and FLOPs remain identical across all settings. Inference latency is reported as mean $\pm$ standard deviation to account for measurement variability, while minor differences in training time per epoch arise from loss-specific computations.

Metric	Adap. Focal	AG-BCE	PT Focal	Dice	BCE
Params (M)	105.28	105.28	105.28	105.28	105.28
FLOPs (G)	49.37	49.37	49.37	49.37	49.37
Train/Epoch (s)	123	118	116	120	117
Infer (ms)	23.0 ± 0.3	23.0 ± 0.3	23.0 ± 0.3	23.0 ± 0.3	23.0 ± 0.3

the performance evaluation of three loss functions — Adaptive Focal Loss, AG-BCE Loss, and PyTorch Focal Loss — on three test images. Each row in the figure corresponds to the segmentation results of one loss function, while the columns represent the respective test images. A legend at the top provides the color coding for the ground truth and the outputs of the three loss functions, allowing for a direct visual comparison of their performance.

5.5. Novelty of the adaptive function

Traditional loss functions, such as PyTorch's Focal Loss, assume equal contributions from all image regions. However, this assumption falls short in medical imaging, where critical areas are harder to segment due to noise, artifacts, or subtle boundaries. The AG-BCE loss function attempts to address this by assigning greater weight to complex regions; however, it does so in a static manner, applying the same weight adjustments across all samples. This static approach fails to capture the dynamic nature of difficulty within and between images.

In contrast, the Adaptive Focal Loss function addresses these challenges by modifying the loss contribution based on the model's forecast confidence, dynamically adjusting the focus to accommodate the unique difficulties of each image or region. This adaptability is achieved by introducing a focusing parameter (γ), which emphasizes hard-to-classify data, and a balancing component (α), which mitigates class imbalance. Adaptive Focal Loss enhances model performance by focusing on the most challenging and clinically relevant areas.

Segmenting prostate borders in micro-ultrasound (micro-US) imaging presents numerous unique challenges. Although micro-US offers superior resolution compared to conventional ultrasonography, it often suffers from severe speckle noise and low contrast, particularly in regions where the prostate boundary is less defined. Variability in tissue density and calcifications further complicates segmentation, as these factors may introduce false signals that confuse the model. Additionally, common artifacts arising from the interaction of sound waves with varying tissue types and densities can distort the true boundaries of the prostate, leading to inaccurate segmentation. Prostate deformation caused by patient movement or pressure effectively handles regions with low contrast, excessive noise, or significant anatomical variations in images, hindering the model's ability to generalize.

To address these challenges, the Adaptive Focal Loss function dynamically adjusts the learning focus based on the unique obstacles posed by each sample. By varying the contribution of each pixel to the loss based on prediction confidence, the model is encouraged to allocate more learning capacity to challenging cases where segmentation is most complex. This approach enables the model to handle regions with low contrast, excessive noise, or significant anatomical variation more effectively. Moreover, the dynamic adjustment of the loss function through the focusing parameter allows the model to tackle artifacts and distortions in micro-US images efficiently. Unlike static methods, this approach recognizes that the importance of different data regions varies depending on the specific conditions of each image. This context-aware strategy enables the model to better interpret the data, thereby improving generalization to unseen cases.

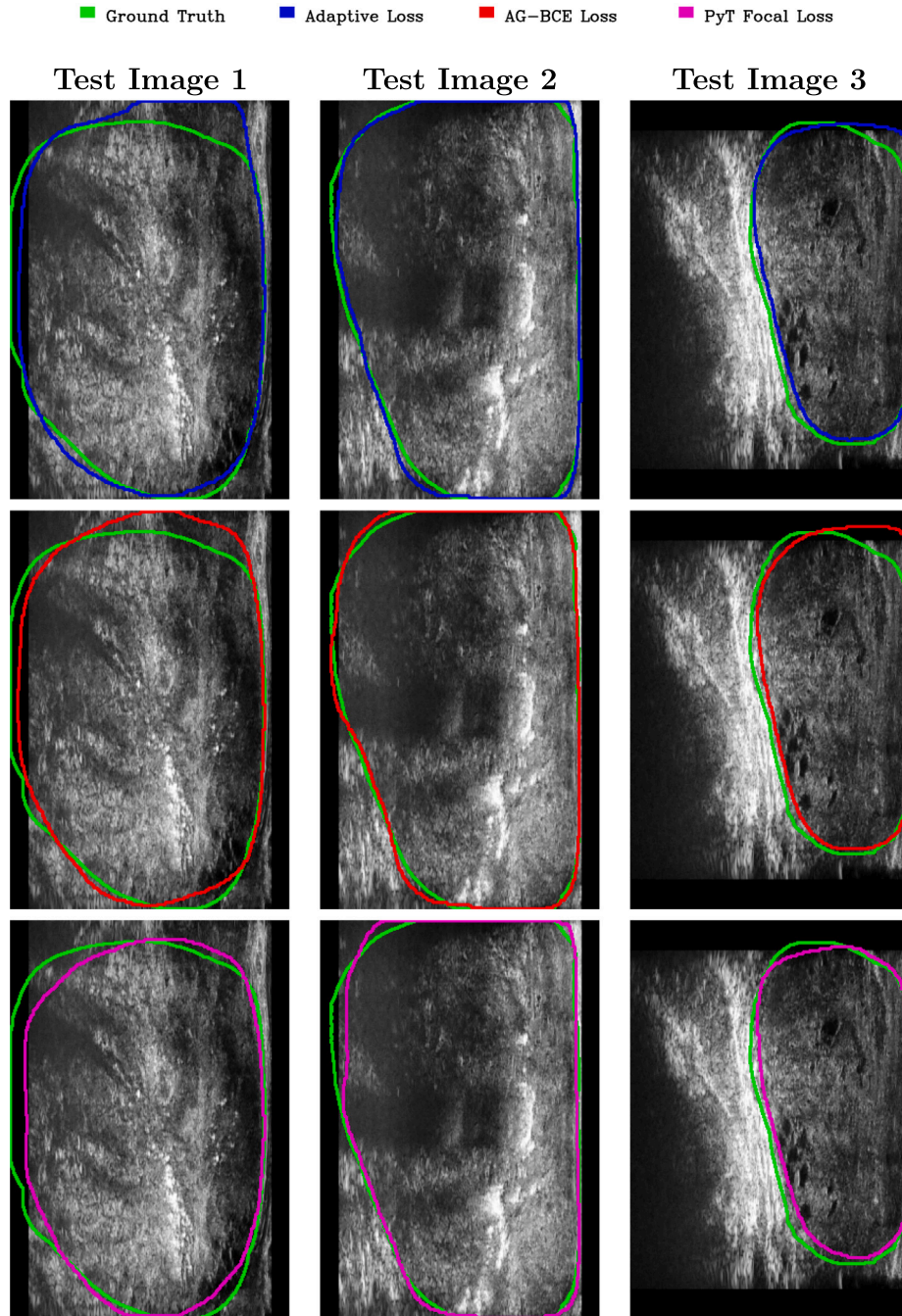


Fig. 4. Segmentation comparison of different loss functions. Segmentation results on three test images using Adaptive Focal Loss, AG-BCE Loss, and PyTorch Focal Loss. Each row shows the outputs for one loss function, and the columns correspond to different test images. The legend at the top indicates the color coding for ground truth and segmentation outputs.

The visual analysis of segmentation results, illustrated in Fig. 5, demonstrates that Adaptive Focal Loss consistently achieves more accurate segmentation, particularly in regions with low contrast, high noise, or complex anatomical variations, closely aligning with the ground truth.

1. **Test Image 04:** The Adaptive Focal Loss function excelled in capturing the prostate boundary, especially in the lower-left region obscured by speckle noise. By comparison, AG-BCE and PyTorch Focal Loss exhibited significant deviations, notably under-segmenting the prostate and failing to depict the boundary accurately.

2. **Test Image 11:** The Adaptive Focal Loss function demonstrated superior performance in the upper-right region, where the prostate boundary was subtly defined against surrounding tissue. AG-BCE failed to capture finer details, producing a less smooth boundary and shifting it inward, while PyTorch Focal Loss over-segmented, extending beyond the prostate.
3. **Test Image 14:** Despite complex artifacts in the lower region, Adaptive Focal Loss maintained a boundary closely resembling the ground truth. AG-BCE exhibited minor mismatches, particularly in the lower-right corner, while PyTorch Focal Loss underperformed, displaying over-segmentation and boundary deviations.

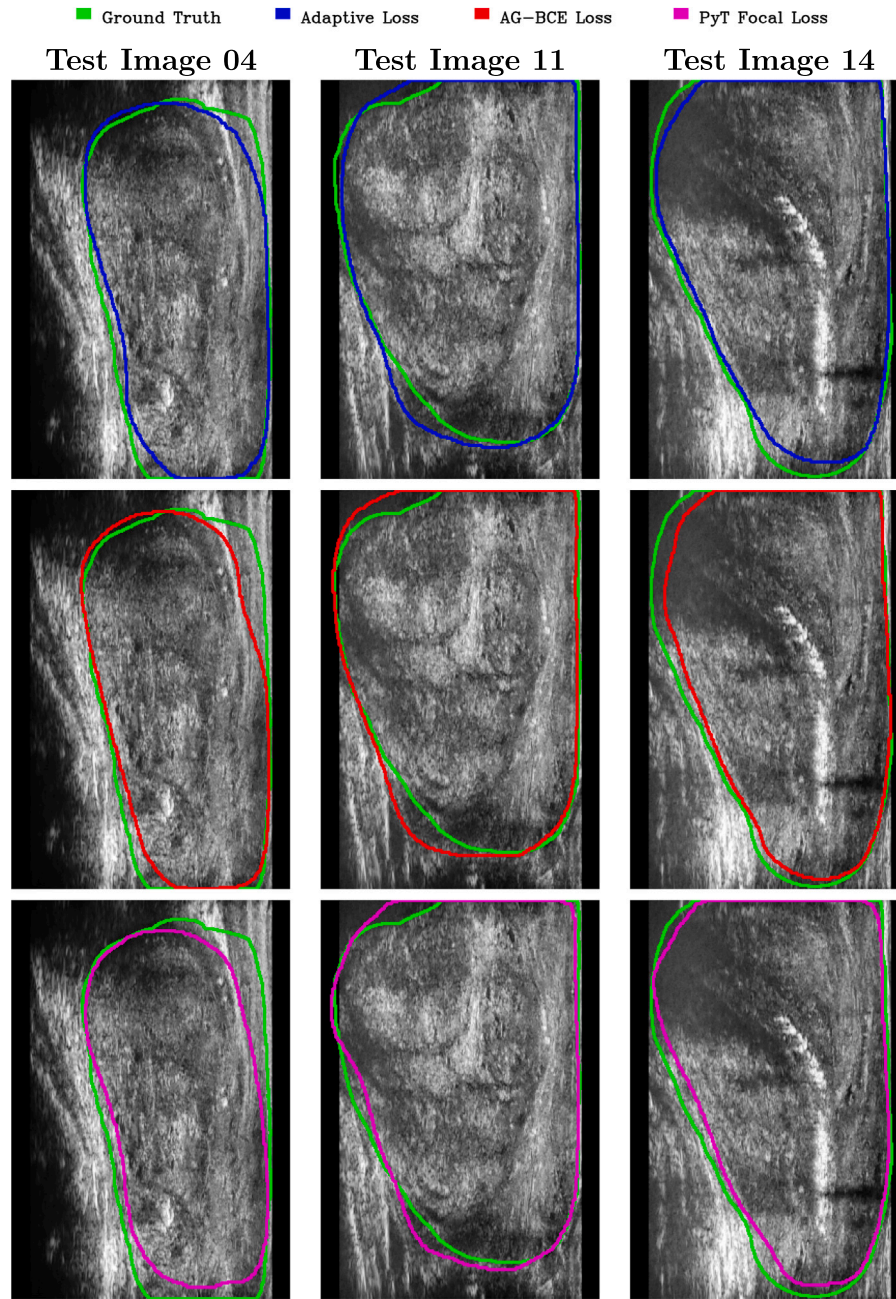


Fig. 5. Comparison of segmentation results on test images 04, 11, and 14. Segmentation results are presented for Adaptive Focal Loss, AG-BCE, and pyT Focal Loss across three test images. Each row corresponds to a different loss function, as indicated in the legend.

Further comparisons of Test 17, Test 18, and Test 19, shown in Fig. 6, underscore the effectiveness of the Adaptive Focal Loss in handling challenging prostate boundary regions.

1. **Test Image 17:** The Adaptive Focal Loss performed exceptionally well in the upper left region, which presented significant anatomical deformation, making segmentation particularly difficult. In contrast, AG-BCE produced an overly smooth boundary, while PyTorch Focal Loss severely under-segmented the area.
2. **Test Image 18:** This image highlighted challenges in the lower prostate boundary caused by artifacts and low contrast. The Adaptive Focal Loss accurately delineated the boundary, whereas AG-BCE exhibited boundary shifts, and PyTorch Focal Loss resulted in under-segmentation.
3. **Test Image 19:** The Adaptive Focal Loss again outperformed the other methods, especially in the upper right region, where

tissue density variations blurred the boundary. AG-BCE over-segmented this region, and PyTorch Focal Loss produced inconsistent results, with instances of both under- and over-segmentation.

Overall, the Adaptive Focal Loss function consistently demonstrated superior performance in accurately delineating prostate boundaries in challenging micro-ultrasound segmentation tasks, outperforming both AG-BCE and PyTorch Focal Loss in accuracy.

5.6. Comparison with state-of-the-art methods on the public micro-ultrasound dataset

Table 4 presents a quantitative comparison between the proposed Adaptive Focal Loss-based method and recent state-of-the-art approaches evaluated on the public Micro-Ultrasound dataset. The

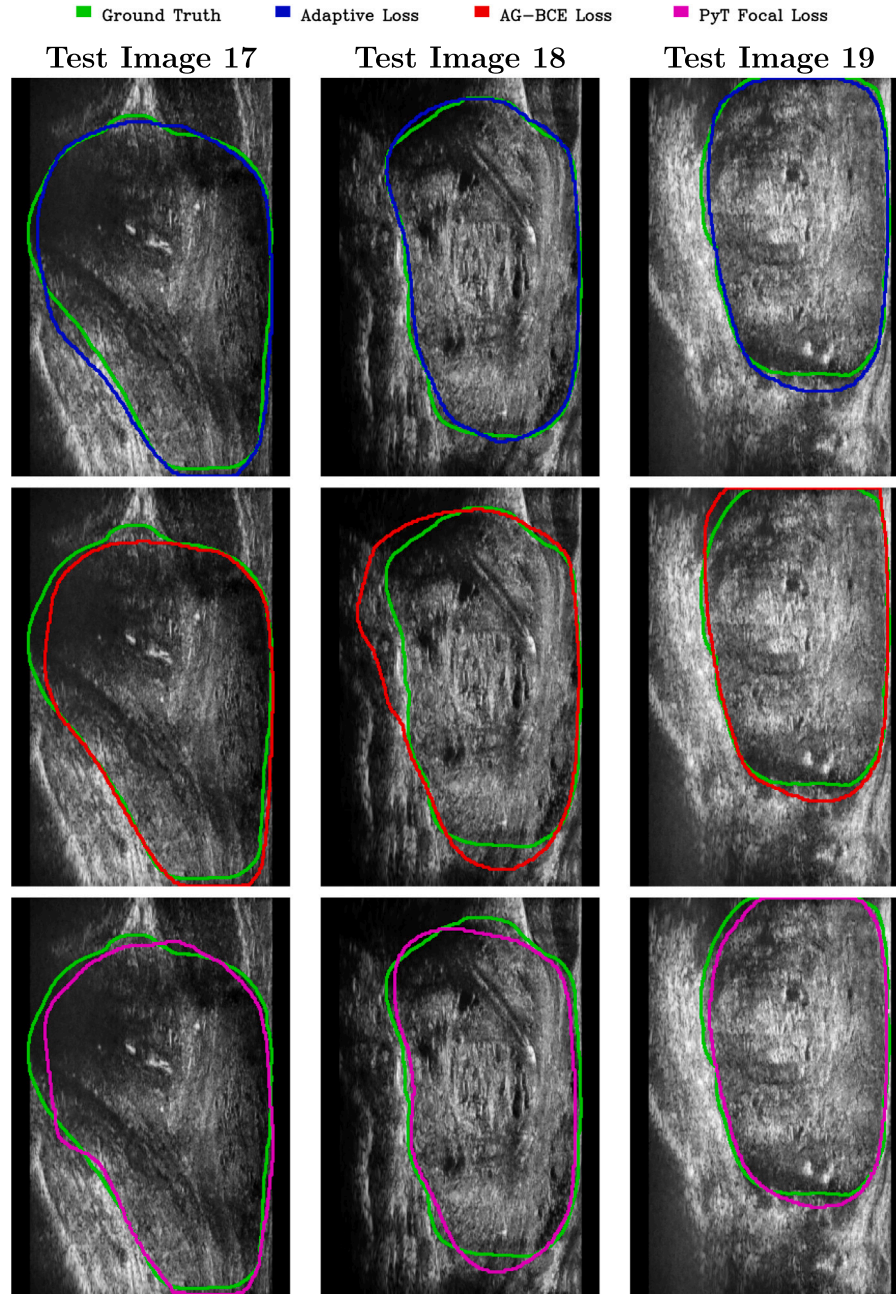


Fig. 6. Comparison of segmentation results on test images 17, 18, and 19. Segmentation results are presented for Adaptive Focal Loss, AG-BCE, and pyT Focal Loss across three test images. Each row corresponds to a different loss function, as indicated in the legend.

proposed method achieves a Dice Similarity Coefficient (DSC) of 0.940 and a Hausdorff Distance (HD95) of 1.949 mm, indicating strong and consistent performance in both region overlap and boundary accuracy. Some existing methods emphasize maximizing region overlap. For example, the proposed DL model (2024) [18] reports the highest DSC of 0.957; however, this improvement in overlap accuracy is not accompanied by a clear advantage in boundary precision, which remains comparable to other methods. In contrast, several approaches demonstrate improved boundary accuracy at the expense of segmentation consistency or rely on complex architectural designs or user interaction. Notably, the 3D micro-US segmentation method (2025) [22] reports a substantially higher HD95 of 2.86 mm, reflecting reduced boundary precision despite achieving a similar DSC.

The main contribution of the proposed method lies in explicitly addressing boundary ambiguity and annotation variability through loss-level adaptation rather than architectural complexity or external guidance. By incorporating Adaptive Focal Loss, the model simultaneously optimizes region-based accuracy and boundary precision, leading to a more balanced and robust segmentation outcome. This balance is particularly important for prostate capsule segmentation in micro-ultrasound images, where boundaries are often ambiguous and noisy. Overall, the proposed approach achieves competitive state-of-the-art performance while providing improved boundary delineation in a principled and computationally efficient manner, making it well-suited for clinically relevant micro-ultrasound segmentation tasks.

Table 4

Comparison with state-of-the-art methods on the public micro-ultrasound dataset.

Method	DSC (↑)	HD/HD95 (mm) (↓)
Proposed DL model (2024) [18]	0.957	1.922 (HD)
MicroSegNet (2024) [10]	0.939	2.02 (HD)
Hypergraph Neural Network (HGNN) (2025) [19]	0.9416	1.93 (HD95)
SAM-based method (2025) [20]	0.940	1.76 (HD)
HEFFLPNet (2025) [21]	0.938	2.12 (HD95)
3D micro-US segmentation (2025) [22]	0.940	2.86 (HD95)
Ours (Adaptive Focal Loss)	0.940	1.949 (HD95)

6. Limitations and future work

While the proposed Adaptive Focal Loss improves segmentation accuracy, particularly in challenging boundary regions, several limitations should be acknowledged. First, the adaptive weighting mechanism introduces additional computational overhead during training due to the dynamic adjustment of weights based on sample difficulty and annotation variability. As quantified in Table 3, this results in a modest increase in training time per epoch compared to simpler loss functions, while the number of model parameters, FLOPs, and inference latency remain unchanged. Consequently, the proposed approach does not affect deployment efficiency, but may be less suitable for resource-constrained training environments or large-scale training scenarios.

In addition, for datasets characterized by low annotation noise or limited inter-annotator variability, the adaptive strategy may yield only marginal improvements over simpler and more computationally efficient loss functions. In such settings, the trade-off between training-time overhead and performance gains should be carefully considered. The method also relies on accurate estimation of sample difficulty; inaccuracies in this estimation may reduce the effectiveness of the adaptive weighting mechanism, whereas loss functions with fixed weighting schemes are less sensitive to such variations.

Another important limitation of the present study is that the evaluation is restricted to a single public micro-ultrasound dataset. This reflects the current state of data availability rather than a limitation of the proposed method. To the best of our knowledge, and consistent with recent work on micro-ultrasound prostate segmentation, this dataset is currently the only publicly available micro-ultrasound dataset providing expert whole-gland prostate annotations together with multiple non-expert annotations, which are essential for explicitly modeling inter-annotator variability [19]. As a result, validation on additional independent micro-ultrasound datasets with comparable acquisition and annotation protocols is not yet feasible.

Future work will therefore focus on extending the evaluation to larger and more diverse micro-ultrasound cohorts as such data become available. In particular, we plan to validate the proposed Adaptive Focal Loss on extended versions of the current dataset and on newly released, multi-institutional expert-annotated micro-ultrasound datasets. These efforts will enable a more comprehensive assessment of generalization within the micro-ultrasound domain while preserving consistency in imaging modality and annotation protocols. In parallel, future research will explore refinements of the adaptive weighting mechanism to improve robustness to extreme cases of sample difficulty and annotation variability, for example, through more stable weighting strategies or meta-learning approaches that adapt loss behavior to dataset characteristics. Addressing these limitations will help establish the Adaptive Focal Loss as a robust and practical loss function for micro-ultrasound prostate segmentation under realistic annotation variability.

7. Conclusion

This work presents an Adaptive Focal Loss framework for prostate capsule segmentation from micro-ultrasound images, specifically designed to address boundary ambiguity, class imbalance, and annotation variability. By dynamically adjusting the loss contribution based on sample difficulty and annotation uncertainty, the proposed method prioritizes clinically challenging regions that are inadequately handled by static loss formulations.

Extensive experiments conducted on a public micro-ultrasound prostate dataset demonstrate the effectiveness of the proposed approach. The Adaptive Focal Loss achieved a mean DSC of 0.940 and a mean HD95 of 1.949 mm on the test set, outperforming standard BCE, Dice Loss, PyTorch Focal Loss, and AG-BCE. In addition, the Adaptive Focal Loss exhibited improved convergence behavior during training, with loss values decreasing from 0.212186 in the first epoch to 0.040419 after ten epochs, indicating stable and effective optimization.

Qualitative analyses further confirm that the proposed loss function produces more accurate and consistent boundary delineation in regions affected by low contrast, speckle noise, and anatomical variability. Compared to AG-BCE and PyTorch Focal Loss, the Adaptive Focal Loss demonstrates superior robustness in difficult prostate boundary regions, resulting in segmentation outputs that align more closely with the ground truth. Unlike approaches that rely on increased architectural complexity or user interaction, the proposed method improves segmentation performance through loss-level adaptation without introducing additional inference-time computational overhead. This makes the framework suitable for real-time clinical deployment and scalable training scenarios.

Overall, the results highlight the effectiveness of adaptive, confidence-aware loss modulation for complex medical image segmentation tasks. The proposed Adaptive Focal Loss offers a principled and extensible solution for micro-ultrasound prostate capsule segmentation, demonstrating strong potential for generalization to other medical imaging applications characterized by severe class imbalance, ambiguous boundaries, and annotation uncertainty.

CRedit authorship contribution statement

Kaniz Fatema: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Vaibhav Thakur:** Writing – original draft, Software, Methodology. **Emad A. Mohammed:** Writing – review & editing, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data utilized in this paper is publicly available from [10]. The code supporting this paper is available at the following GitHub repository [35].

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